Autonomous Power Routing in Stochastic Smart Grids

Three-presenter poster script · Vatsal Bharati, Samarth Mujumdar, Krish Jariwala
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How to use this script. The poster is read in three vertical columns, left to right. Each presenter owns one column, so the talk moves across the poster in the same order the eye does — nobody has to reach across anyone else. Target length is **7 minutes** for the full walk-through; a **90-second** version is on the last page for visitors who are passing by. Lines in the indented blocks are what you say; the amber lines are stage directions (where to point, when to pause, when to hand over).

Division of the poster

Presenter	Poster region	Owns these questions	Time
Vatsal	Title, KPI strip, Abstract, Introduction, Stochastic Simulator, Experimental Setup (left column)	Why this problem is hard; what the simulator is; why the numbers can be trusted	2:15
Samarth	Methodology, Results · Forecast, Results · Decisions (centre column)	How both controllers work; the LSTM forecaster; decision quality vs. the oracle	2:30
Krish	Results · Control, Decentralized Check, Conclusion, References (right column)	The headline result, the honest caveats, what comes next	2:15

PART 1 — VATSAL · The problem and the testbed (2:15)

Opening (stand at the left edge of the poster)

Make eye contact first, then gesture at the title. Do not start reading the abstract aloud.

"Hi — this is our project on autonomous power routing in smart grids. I'm Vatsal, this is Samarth and Krish. Give us about seven minutes and we'll take you across the whole poster; stop us any time.

Here's the problem in one sentence. Every substation on a grid has generation on one side and demand on the other, and the two never match. With rooftop solar and wind in the mix, the gap — we call it the **imbalance** — flips between surplus and deficit several times a day. An operator has three ways to close that gap: curtail generation, shed load, or **route power** to a neighbour who is in the opposite condition. Routing is by far the cheapest, so that's the decision we're trying to automate."

Why it isn't easy

Point at the Introduction box, then at Fig. 1 (the A–B–C triangle).

"The catch is that the decision isn't local. A transfer that looks sensible at one node can overload a line somewhere else, so the controller has to decide **where** power goes, **how much**, and at **what loss and risk**. That's a sequential decision problem, which is why we framed it as reinforcement learning rather than as a rule.

We asked four questions — and I'll be honest up front that we answer the first two properly, the third partially, and the fourth not at all. How much imbalance can a centralized learner remove, measured against a real upper bound? How does a decentralized, graph-based controller behave on the same grid? Can a safety cost keep that decentralized learning inside physical limits? And would it transfer to a bigger topology unchanged — that last one is future work."

The simulator

Hand flat over the Stochastic Simulator panel; trace the triangle in Fig. 1 while you talk.

"Everything runs on a simulator we wrote ourselves — three nodes: A is a city, B is a suburb, C is industrial. It is deliberately not a toy. Load and generation follow diurnal cycles with different phases per node, weather is a three-state Markov chain — normal, heatwave, storm — that rescales solar and wind, the noise is an AR(1) random walk rather than white noise so that errors are autocorrelated the way real forecast errors are, and we inject rare events: solar outages at A, brown-outs at B, industrial demand spikes at C.

On top of that sits the physics — a DC power-flow model with impedance, I-squared-R losses and thermal limits, slew-rate-limited actuators, and N-minus-1 contingencies, which is this dashed line here: we pull a line out and the controller has to cope.

And we checked that the signal is actually what we claim. Lag-one autocorrelation is 0.95 with a clear 24-hour echo, a t-test rejects zero-mean imbalance at all three nodes, and a Levene test confirms the weather regimes really do change the variance. That's what justifies the six-hour observation window you'll see in a moment."

How the numbers are kept honest

Point at Experimental Setup, then up at the KPI strip. Slow down here — this is the paragraph that earns the headline number.

"One thing we care about more than the result itself: a percentage means nothing without a reference. So every controller is scored on the same **240 unseen hours**, generated from a different seed than anything it trained on, against an unmanaged baseline of 16,303 megawatts of accumulated imbalance.

And we built a **greedy oracle** — at every hour it brute-forces all seven actions and takes the best one. That's the ceiling for one 20 MW transfer per hour. We also report always-Hold and random, because on this task doing nothing is a legal move and doing something arbitrary is actively harmful. Those four reference points are what let Krish say anything meaningful at the end.

Samarth will take you through what's actually inside the controllers."

Handoff: *step back and to the left so Samarth can own the centre column.*

PART 2 — SAMARTH · The two architectures and what they learned (2:30)

The centralized controller

Start at the top of the Methodology panel and move down it box by box — the arrows do the explaining for you, follow them with your finger.

"Thanks. We built two controllers deliberately representing the two architectures the literature argues about, and we put them on one simulator with one evaluation harness — that's really the contribution.

The first one is **centralized**. It sees a six-hour window of imbalance at all three nodes. That goes into a Transformer encoder — four attention heads, key dimension 32, with a learned positional embedding. We use attention rather than a plain recurrent net because the signal has both a short-lag component and a 24-hour component, and attention lets the controller weight hours by relevance instead of by recency.

On top of that is a standard DQN head — replay buffer, separate target network — emitting Q-values for seven actions: hold, or move 20 MW along one of the six directed node pairs.

The reward is the part worth pausing on. It's **differential**: total absolute imbalance before the action minus after, less a cost on the I-squared-R loss. In other words, the megawatts of grid stress the action actually removed. We chose that over rewarding absolute imbalance because the pre-action total is a constant of the state — a big state-dependent offset that swamps the Q-targets and teaches the agent nothing about which action to pick."

The decentralized controller

Move to the lower half of Methodology (the green-tagged block).

"The second is **decentralized**, and it throws away the central view on purpose. One agent per substation. Each one sees only its own imbalance, the hour, what its neighbours broadcast, and the live adjacency matrix — so it knows the topology, including when a line drops out. A graph neural network does the neighbour aggregation, which is what would let this scale to a grid with more than three nodes.

The actuators are continuous here, not discrete: a set-point in plus-or-minus 50 MW with a 15 MW-per-hour slew limit and a 0.5 actuator response, so it can't teleport a line to capacity in one step. We train it two ways — plain PPO, and CPO, which is PPO with a hard safety constraint rather than a penalty term. Krish will tell you where that stands."

Result 1 — the forecaster

Point at Fig. 2, the two horizontal bars.

"Two results from my column. First, a sanity check that the six-hour window carries real information. We pre-trained an LSTM to predict next-hour imbalance. R-squared is 0.911, and held-out RMSE is 7.54 megawatts against 9.57 for a persistence baseline that just repeats the last value — a 21 percent error reduction. Note the split is **chronological**, not random: the windows overlap by five of their six hours, so a shuffled split would leak nearly identical windows across the cut and inflate that number."

Result 2 — decision quality

Point at the three stat tiles (59.8% / 0.488 / 100%). The last sentence is the important one.

"Second: how often does the controller choose the same action the oracle would? Just under 60 percent of hours. But raw agreement is a weak number on this task, because Hold dominates the labels — so we report Cohen's kappa, which corrects for chance agreement: 0.488, where an always-Hold policy scores exactly zero. And hold is only 39 percent of oracle labels, so the agent is not scoring by doing nothing.

The 100 percent is a correctness check, not a performance claim: our oracle and our environment are two independent implementations of the same physics, and we verified they agree on every single test state. If they didn't, every other number on this poster would be scoring the agent against a different grid from the one it trained on.

Krish — the physical result."

Handoff: *Samarth steps left, Krish moves to the right column.*

PART 3 — KRISH · Results, caveats and conclusion (2:15)

The headline

Point at Fig. 3. Walk down the bar chart — oracle first, then the agent — so the ceiling lands before the result.

"So, the number we actually care about: total imbalance over those 240 unseen hours.

Read the chart from the top. The greedy oracle removes **24.38 percent** — that is the ceiling, nothing with a single 20 MW transfer per hour can beat it. Our Transformer-DQN removes **23.62 percent**. That's **96.9 percent of the attainable control**, and that framing is the whole point: 23.62 on its own could mean anything, and against a ceiling of 24.38 it means the controller is close to done.

Below it, the transfer-learning variant — the same DQN over a frozen LSTM encoder — reaches 22 percent. Always-Hold is exactly zero by construction, which is our harness sanity check. And random is **minus 15 percent**: acting arbitrarily is substantially worse than not acting at all. That's worth a second. It means this is not a task where any transfer helps, so beating zero is a real result rather than an artefact."

The cost

Point at the amber trade-off box.

"It isn't free, and we didn't hide it. The agent switches more often than the oracle and burns more I-squared-R loss to get there. Imbalance reduction and routing efficiency are two different objectives, and we optimised the

first. A system operator would want both in the reward."

The honest caveat

Point at the Decentralized Check table. Say this plainly — do not bury it. Evaluators reward teams who state their own limits before being asked.

"Now the part we want to be straight about. The decentralized side is **verified as an implementation, not as a result**. What we can stand behind: the environment, the observations, the actuators and the safety machinery all work. Across five CPO seeds and 150 episodes we recorded **zero safety cost and zero thermal or voltage violations** — the constraint does what it's supposed to do. And a classical proportional rule on the same 48-hour reference removes 21.24 percent for 1.32 MW of loss, which gives us the reference the learned controller will have to beat.

What we cannot yet claim is that the learned decentralized controller is any good. The PPO-versus-CPO comparison isn't finished, so there is no decentralized performance number on this poster. We'd rather show you an empty cell than a number we can't defend."

Close

Step back half a pace and take in the whole poster — the close should feel like it covers all three columns, not just yours.

"Four things to take away. One: a Transformer-DQN recovers almost 97 percent of the attainable control on a stochastic three-node grid. Two: it isn't free — reduction and efficiency are distinct objectives, and that's a finding, not a footnote. Three: the graph multi-agent side has a verified foundation with a working safety constraint, but no performance claim yet. Four: this is synthetic data, and the next steps are the PPO versus CPO comparison, double-Q learning to take the edge off value overestimation, and a larger topology — which is where decentralization should start paying for itself.

The honest version of our conclusion is that the centralized result is solid and the decentralized one is unfinished. Happy to take questions."

QUESTION BANK

Answers are assigned to whoever owns that part of the poster, so the team isn't talking over itself. If a question lands on someone else's column, say "that's Samarth's" and hand it over — that reads as coordination, not evasion.

Question	Who	Answer
Is 23.62% good? It sounds small.	Krish	That is the right instinct, and it's why the oracle bar is there. The maximum any policy can remove with one 20 MW transfer per hour on this window is 24.38%, so we are at 96.9% of what is physically reachable. The remaining headroom is 0.76 percentage points, not 76.
Why only three nodes?	Vatsal	Because the point was a controlled comparison, not scale. Three nodes let us compute an exact greedy oracle by brute force — seven actions per hour — which is what makes every percentage interpretable. At 50 nodes there is no tractable oracle and you are back to uninterpretable percentages. Scaling is RQ4 and it is future work.
Why a Transformer for a 6-step window? Isn't that overkill?	Samarth	Fair challenge. The justification is in the autocorrelation: lag-1 is 0.95 and lag-24 is 0.87, so the signal has two timescales, and attention weights hours by relevance rather than recency. We also ran the comparison — the frozen-LSTM transfer agent on the identical task reaches 22.0% against the Transformer's 23.62%.

Why did transfer learning do *worse*? Isn't pre-training supposed to help?	Samarth	That's our favourite negative result. The encoder was pre-trained to forecast next-hour imbalance, which is a different objective from ranking seven actions by their immediate effect. The features that predict where imbalance is going aren't the features that discriminate between actions, and with the encoder frozen the agent can only learn a small head on top. The measured cost of that constraint is about 1.5 percentage points.
Your data is synthetic. Why should I believe any of it?	Vatsal	You shouldn't believe it transfers to a real network — we say that in the limitations. What synthetic data buys us is a known ground truth: we can compute the true optimum, generate unlimited unseen hours and re-run everything from one seed. We validated that the signal has the statistical properties of a real grid signal (diurnal period, autocorrelation, heteroscedasticity by weather regime) rather than assuming it.
What happens when a line goes out?	Vatsal	That's the N-1 contingency in Fig. 1. The topology is fed to the controller live — for the decentralized agents as an adjacency matrix — so an outage changes the observation rather than silently breaking the assumptions.
Why isn't there a decentralized performance number?	Krish	Because the PPO-versus-CPO evaluation isn't complete. We have the environment, the safety result (zero violations over five seeds) and the proportional-rule reference, but not a defensible learned-controller number. We chose to leave the cell empty rather than report a run we'd have to retract.
Would the decentralized version ever beat the centralized one?	Krish	Not on three nodes — the centralized controller sees everything and the oracle is almost reachable, so there is nothing to win. The case for decentralization is scale and no single point of failure, and testing that needs a bigger topology. That's exactly RQ4.
How do you know the agent isn't just learning “always hold”?	Samarth	Three ways. Hold is only 39% of oracle labels. Cohen's kappa is 0.488 where always-Hold scores exactly 0.000 by construction. And always-Hold removes 0.00% of imbalance while the agent removes 23.62%.
Is this real-time capable?	Samarth	Inference is a single forward pass over a 6×3 window — microseconds — and the control interval is an hour, so compute isn't the constraint. What is unproven is everything else: real telemetry, state estimation, and operator trust in a learned policy.
What would you do with another month?	Krish	Finish the PPO/CPO comparison, repeat every run over multiple seeds so we can report confidence intervals instead of single runs, and scale past three nodes. In that order.
Why DQN and not a policy-gradient method centrally?	Samarth	The centralized action space is seven discrete actions, which is DQN's natural setting; PPO is on the decentralized side because those actuators are continuous set-points. The architecture follows the action space.

90-SECOND VERSION

For visitors who clearly won't stay. One speaker delivers it — whoever is free — and the other two stay quiet. Finish with the offer; don't trail off.

“Thirty-second version: grids have to be rebalanced every hour, and routing power between substations is the cheapest way to do it. We built a stochastic three-node grid simulator with real physics — power flow, losses, thermal limits, line outages — and trained a Transformer-based deep Q-network to decide where to send power each hour.

Over 240 unseen hours it removes 23.6 percent of total imbalance. The reason we can tell you that's good is this bar *(point at the oracle)*: we brute-forced the optimal action every hour, and the ceiling is 24.4 percent. So we're at 96.9 percent of what's physically achievable.

We also built a decentralized version — one agent per substation, graph neural network, safety-constrained — and that one we're only claiming as a verified implementation: zero safety violations over five seeds, but the

performance comparison isn't finished.

If you want the detail, any of the three of us can walk you through the full poster — it's about seven minutes."

DELIVERY NOTES

Rehearse these four

- **The handoffs.** Two seconds of silence while someone shuffles across is the only thing an audience reliably remembers. Each handoff has a scripted last line — learn those three sentences cold even if you paraphrase everything else.
- **"96.9% of the ceiling."** This is the sentence the whole poster exists to support. Whoever is asked, the answer starts with the oracle, not with 23.62.
- **The caveat.** Krish states the incomplete PPO/CPO work without hedging or apologising. Stated plainly it reads as rigour; mumbled it reads as something you hoped nobody would notice.
- **Pointing.** Point at the figure *before* you describe it, and keep your body turned towards the listener, not the board.

Say this, not that

Don't say	Say
"It's only a 23% improvement."	"It recovers 96.9% of the attainable control."
"The decentralized one didn't work."	"The decentralized implementation is verified; the performance evaluation isn't finished."
"It's just simulated data."	"It's a synthetic grid, which is what gives us a computable optimum to measure against."
"I think / I'm not sure but…"	"We didn't measure that — here's what we did measure."
Guessing at a number you don't remember	"It's in the report; let me not guess at it."

Before you present

- Agree who takes the first question if it's ambiguous — default to the column it came from.
- Decide your fallback if only one of you is at the board when a visitor arrives: give the 90-second version, don't wait.
- Keep the full report open on a laptop or phone for exact figures (per-class F1, training curves, telemetry tables); don't put those numbers on the poster or in your head.

One thing to settle before you present. The figures on the poster differ slightly from those in `PBL_Project_Report.docx` — the poster reports 23.62% reduction, a 24.38% oracle ceiling and $\kappa = 0.488$, while the report states 23.52%, 24.39% and $\kappa = 0.586$. The decentralized sections differ more substantially: the report records a completed PPO evaluation at −14.93% against a 21.71% proportional rule, whereas the poster describes the PPO/CPO evaluation as incomplete. Confirm which run is current and make sure all three of you quote the same set of numbers — a visitor who has seen both documents will notice, and the correct answer ("the poster reflects a later re-run") only sounds convincing if you give it immediately.